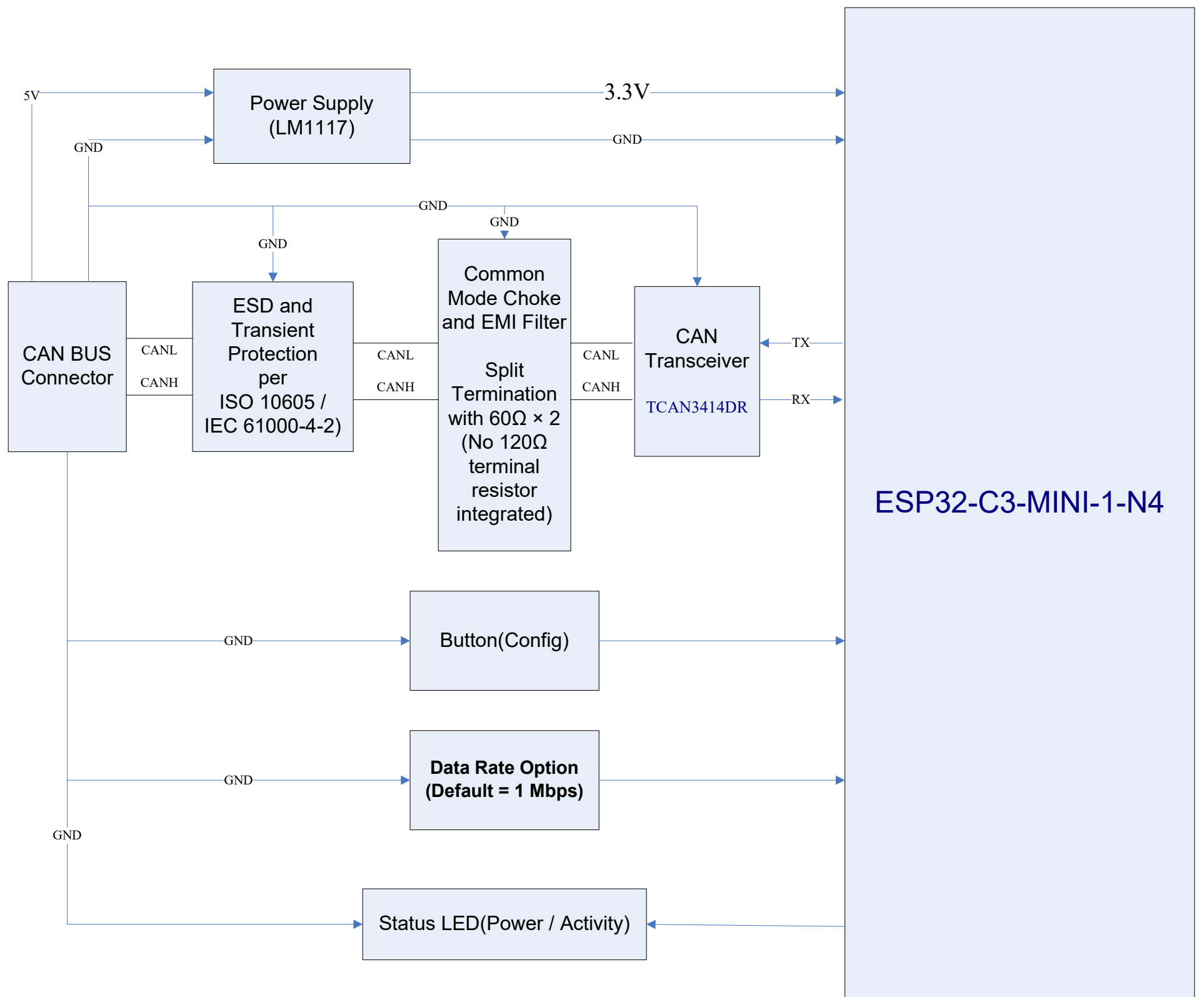


CONV-CAN(CAN-WiFi-003)



The CONV-CAN (CAN-WiFi-003) is a CAN-to-WiFi bridge module based on the ESP32-C3-MINI-1-N4 microcontroller. External 5V power is regulated to 3.3V by an LM1117 LDO to supply the ESP32 module and CAN transceiver. The CAN bus interface includes:

A standard CAN bus connector for CANH, CANL, and GND.

ESD and transient protection per ISO 10605 / IEC 61000-4-2.

A common mode choke combined with EMI filtering and split termination ($60\Omega \times 2$), with no integrated 120 Ω terminal resistor.

A 3.3V CAN transceiver (TCAN3414DR) interfacing TX/RX signals to the ESP32.

User configuration is supported via a configuration button for data rate selection (Default = 1 Mbps) and status indication through LEDs (Power/Activity).

This design ensures robust EMI performance and compliance with FCC Part 15 Class B requirements.